

06. Selection Process Analysis

One-line summary

Candidate groups produced through candidate generation and nested reference may undergo additional selection, reordering, and candidate-reduction processes before the final arrangement is displayed.

Within the observations covered here, both RF4SP and RF5 show candidate-order changes, relatively fixed behavior under some conditions, lower success rates as Candidate Count increases, instability after the candidate group exceeds three, and possible differences in the priority of Auto-Arrange candidates.

This chapter organizes those observations around two ideas: **Candidate Count N** and **candidate priority**. The internal random-number algorithm, weighting, and exact processing order remain unresolved. Terms such as “selection,” “reselection,” and “candidate reduction” are convenient observation labels and do not prove the internal implementation.

6-1. Processing After Candidate-Group Generation

One-line summary

The materials generated as candidates may receive further processing rather than being registered directly as the final arrangement.

Chapters 4 and 5 described how candidates not directly inserted by the player may enter the group through performance-source contamination, internal arrangement, and nested reference. Completed equipment, however, also shows:

- changes in candidate order;
- changes in displayed order;
- differences between RF4SP and RF5; and
- cases in which an Auto-Arrange candidate appears to be displaced.

These observations are consistent with one or more later-stage processes, such as reordering, selection, candidate reduction, priority handling, or reselection. In RF5, conditions with three or fewer candidates often behave relatively consistently, while conditions with four or more candidates become less stable.

When an existing arrangement is present, there are also cases in which an Auto-Arrange candidate is not observed.

Accordingly, **candidate-group generation** and **final arrangement determination** may be separate stages. The boundary between those stages is not directly observed.

6-2. Candidate Count and Success Rate

One-line summary

Increasing Candidate Count may reduce the chance that the intended materials remain in the three final arrangement slots.

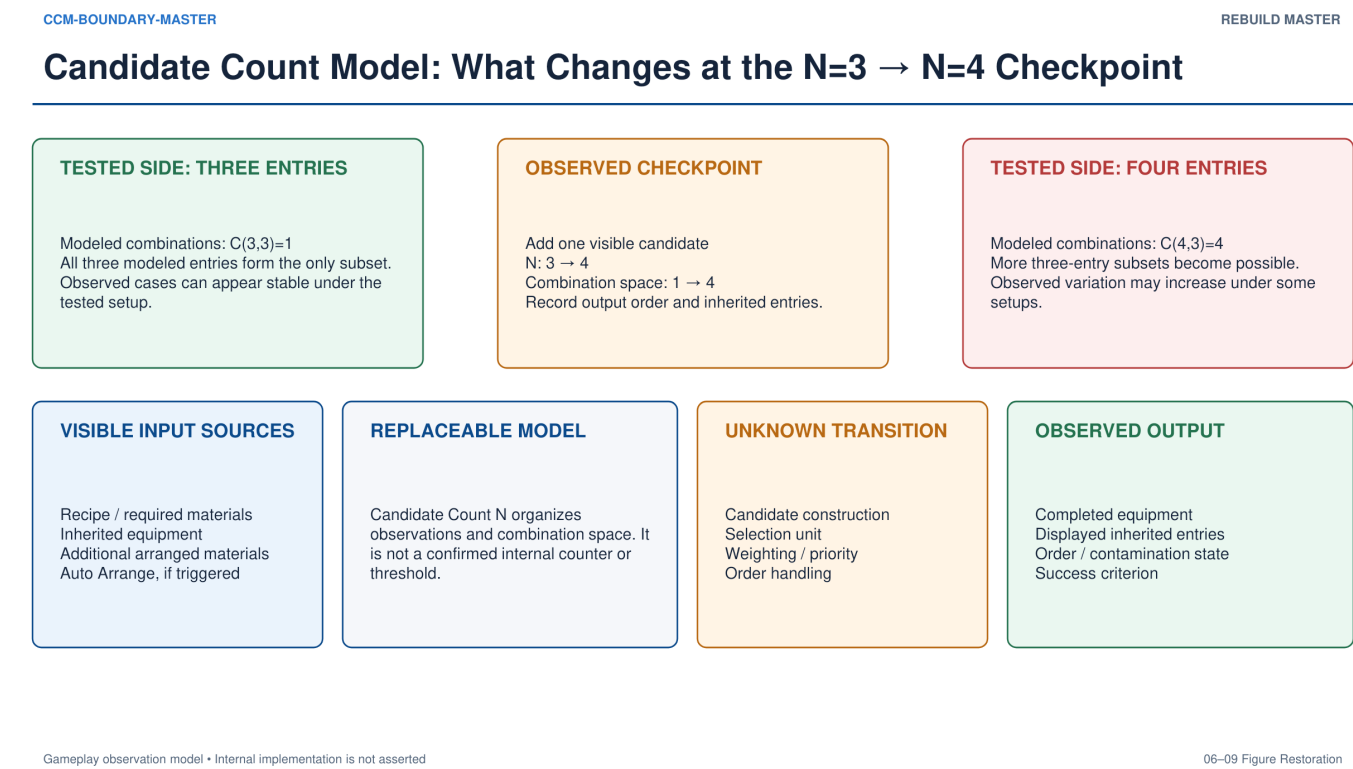
Candidate generation, performance-source contamination, nested reference, and Auto-Arrange can increase the size of the candidate pool. Possible candidates include:

- the source equipment;
- its internal arrangement;
- performance-source contamination candidates; and
- Auto-Arrange candidates.

When three distinct intended materials must be selected from a candidate pool of size N, a simple equal-selection approximation is:

Figure 06-P02-img-000

Master: CCM-BOUNDARY-MASTER | Source anchor: Japanese Canonical p.2

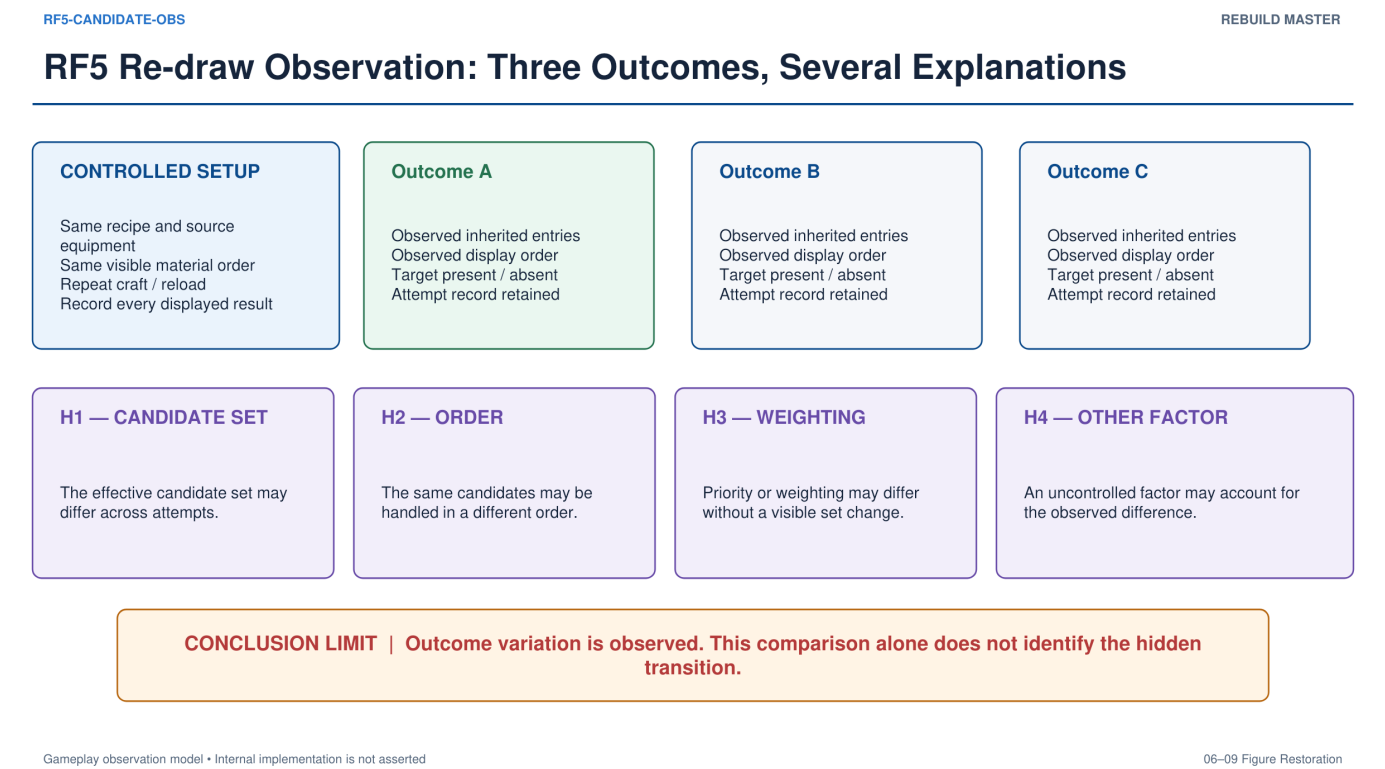


Placement caption

Candidate generation / draw-processing concept map
Evidence boundary: the figure separates Observation, Model, and Unknown Transition as applicable.

Figure 06-P03-img-001

Master: RF5-CANDIDATE-OBS | Source anchor: Japanese Canonical p.3



Placement caption

RF5 re-draw observation comparison
Evidence boundary: the figure separates Observation, Model, and Unknown Transition as applicable.

$$P = 1 / C(N, 3)$$

This is a combination model, not a proven description of the game's random-selection implementation.

Messhilite inheritance and Glitter Augite arrangement can coexist. In observed cases, crafting with one Messhilite Ore and three Glitter Augites could produce an item in which all three Glitter Augites occupied the arrangement slots while the Messhilite performance inheritance still succeeded.

Required base materials	Additional Glitter Augites possible	Candidate pool N	Approximate success rate
1	3	7	57.1%
2	2	6	50%
3	1	5	40%
4	0	4	25%
5	Not possible	-	Messhilite inheritance unavailable

The table is a simplified expectation for a three-distinct-material target and assumes equal selection.

Repeated copies of the same intended material increase the number of winning combinations, so a larger candidate pool is not necessarily disadvantageous in every construction.

Candidate Count may also affect the number of trials, resets, and the efficiency of split-save procedures. Candidate weighting, equal-selection behavior, and priority handling were unresolved in the Japanese Canonical source.

Post-Canonical observation: a boundary on the equal-selection assumption

An additional RF4SP test examined four distinct candidates competing for three final slots. Across 50 trials, the four possible candidate subsets occurred as follows:

Candidate subset	Observed count	Observed rate	Equal-selection expectation
Turnip / Pink Turnip / Golden Turnip	0	0%	12.5
Turnip / Turnip Seed / Pink Turnip	5	10%	12.5
Turnip / Turnip Seed / Golden Turnip	42	84%	12.5
Turnip Seed / Pink Turnip / Golden Turnip	3	6%	12.5

Goodness-of-fit test against equal 25% subset selection: $\chi^2 = 93.84$, $df = 3$, Cramér's $V \approx 0.791$. The equal-subset hypothesis is rejected.

This result does **not** by itself invalidate Candidate Count as a model for organizing the combination space. It shows that the auxiliary assumption that every candidate subset is selected with equal probability may have exceptions. The cause remains **Unknown**: material-specific properties, registration order, candidate-generation order, or another internal process may be involved.

The initial 30-trial imbalance remained at a similar level after 20 additional trials. Replication with other material sets is still required before generalizing the result.

6-3. Randomness in RF4SP

One-line summary

RF4SP may show order variation even at a relatively early stage of the inheritance process.

RF4SP accessory inheritance showed candidate-order changes, displayed-order changes, and order indeterminacy. In Golden Bracelet-related inheritance, Cheap Bracelet, Bronze Bracelet, and Silver Bracelet appeared in different orders under closely related conditions.

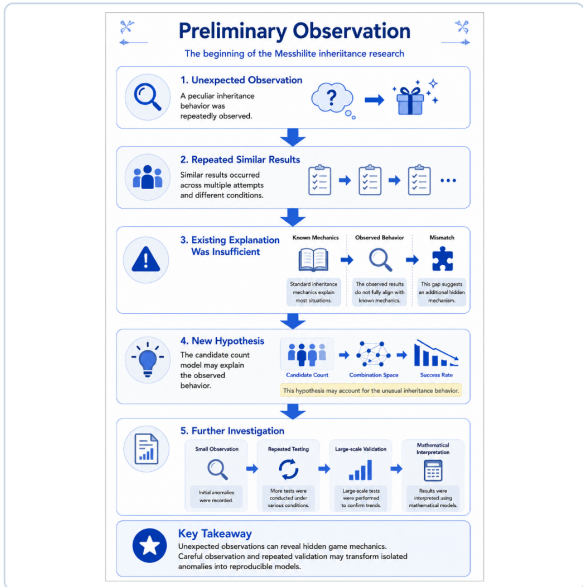
Figure 06-P05-img-002

Master: MESSHILITE-PRELIMINARY-OBS | Source anchor: Japanese Canonical p.5

MESSHILITE-PRELIMINARY-OBS

REUSE / TRANSLATE

Messhilite Preliminary Observation



EXPLANATORY FUNCTION RETAINED

The source figure retains the input materials, crafting record, displayed result, preliminary comparison, and follow-up question.

CURRENT INTERPRETATION

Preserve the observed input/output record. The preliminary difference motivates validation but does not identify the causal transition.

SOURCE DETAIL + CURRENT BOUNDARY

Gameplay observation model • Internal implementation is not asserted

06–09 Figure Restoration

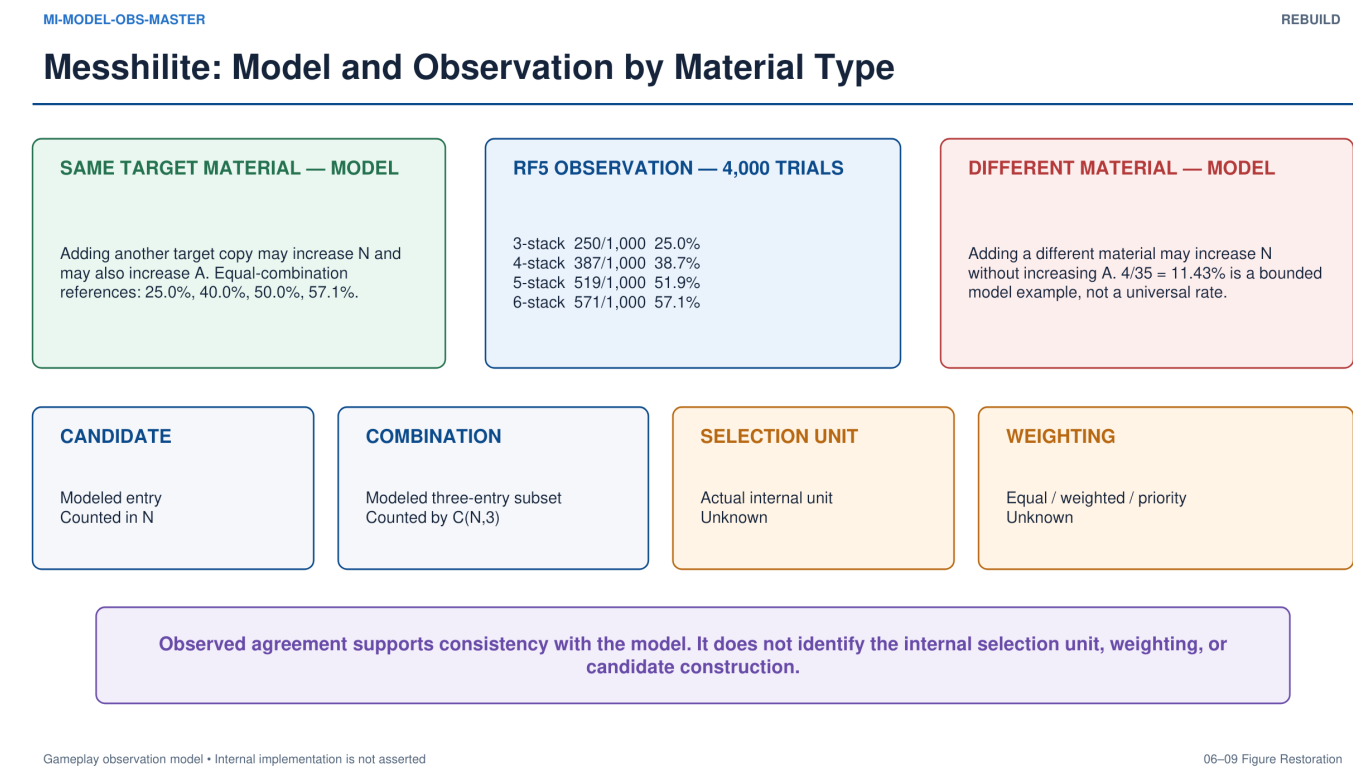
Placement caption

Messhilite + Glitter in-game observation record

Evidence boundary: the figure separates Observation, Model, and Unknown Transition as applicable.

Figure 06-P07-img-004

Master: MI-MODEL-OBS-MASTER | Source anchor: Japanese Canonical p.7



These observations are consistent with an internal shuffle or displayed-order redetermination occurring relatively early. However, they do not establish complete randomness, equal probability, or a fixed internal priority.

Post-Canonical observation: Object X position test

An RF4SP test used the same three arrangement entries in every trial: two Object X effects reversed by Object X and two Mealy Apples. Across 300 trials, the final position of Object X was:

Final position of Object X	Observed count	Rate	95% Wilson interval
First	163	54.33%	48.68%-59.88%
Second	92	30.67%	25.72%-36.10%
Third	45	15.00%	11.40%-19.48%

Under an equal-position hypothesis, each position would be expected 100 times. The observed result gives $\chi^2 = 70.58$; the equal-position hypothesis is rejected ($p < 0.000001$).

This test concerns **the final order of three already-present entries**. It is not the same question as which three candidates survive when Candidate Count exceeds three. The cause of the position bias remains unresolved.

6-4. Relatively Fixed Behavior in RF5

One-line summary

RF5 may behave relatively consistently while Candidate Count remains three or fewer.

Within the observed RF5 conditions, three or fewer candidates often produced repeated results, relatively fixed displayed order, and stable outcomes. When Candidate Count reached four or more, result changes, order changes, and candidate instability became more visible.

When existing arrangement slots were already occupied, some tests did not show an Auto-Arrange candidate entering the final arrangement. This may indicate priority differences among candidate types, but it does not establish a fixed priority algorithm.

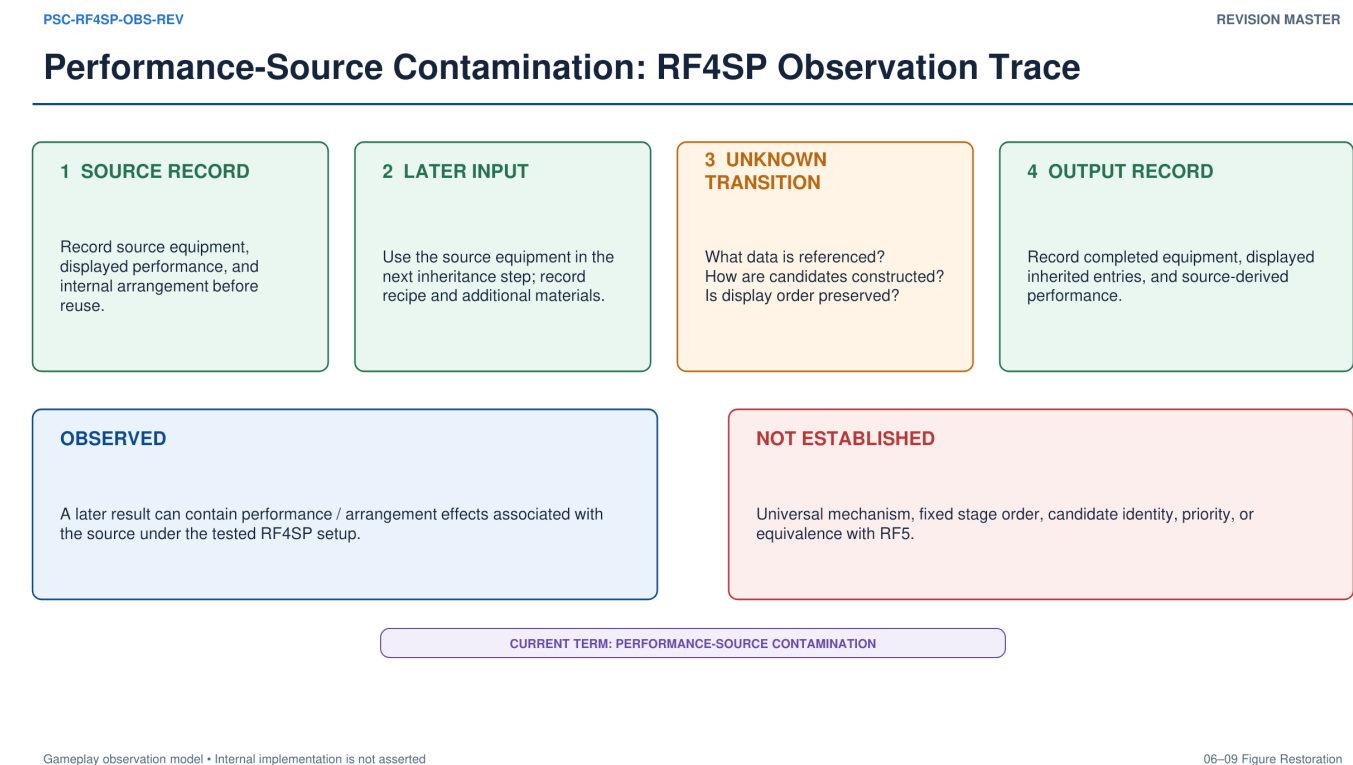
Candidate type	Observed tendency	Provisional priority	Boundary
Existing arrangement	May be retained	High	Condition-dependent
Internal candidates of performance source	Nested expansion observed	High	Special reference remains possible
Ordinary added material	Usually becomes a candidate	Medium	Basic candidate group
Performance-source contamination candidate	Appears conditionally	Medium	Trigger unresolved
Auto-Arrange candidate	More likely when a slot is open	Low to medium	Possible displacement
Internal material inside Auto-Arrange candidate	Expansion not observed in some cases	Low	May be outside nested reference
RF4SP order-indeterminate candidate	Display order varies	Unresolved	Display-order processing unresolved

This table is an observation-based provisional organization. Candidate Count, open slots, inheritance position, and candidate type may all contribute.

Comparison point	RF4SP	RF5
Early order indeterminacy	Observed relatively early	Relatively fixed with small candidate groups
Candidate Count ≤ 3	May still show order variation	Relatively stable in observed conditions
Candidate Count ≥ 4	Variation may continue	Instability becomes more visible
Display-order changes	Observed	Observed conditionally
Performance-source contamination	Observed	Observed

Figure 06-P09-img-005

Master: PSC-RF4SP-OBS-REV | Source anchor: Japanese Canonical p.9



Placement caption

RF4SP Performance-Source Contamination causal diagram
Evidence boundary: the figure separates Observation, Model, and Unknown Transition as applicable.

Auto Arrange Boundary Condition Model

Observation-Based Boundary Condition Model

Objective

Verify the boundary condition under which
Auto Arrange (generated from required materials)
remains in the interference candidate pool.

1. Test Flow

Base Equipment (Base requires 4 elements) 	+	Performance Source (Condition A+C) 	+	Additional Arrangement Material (Subunit) 	+	Final Symbols Check whether Auto Arrange interference candidates.
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2. Comparison Conditions and Results

Condition A No existing arrangement (Auto Arrange has been in support) Observed Result Auto Arrange Leather Boots Formula A Observation: Auto Arrange required.	Condition B One existing arrangement (Total candidates = 0) Observed Result Leather Boots Leather Boots Formula A Observation: Auto Arrange not observed. Order remained stable.	Condition C Two existing arrangements (Auto Arrange no longer remained) Observed Result Formula A Leather Boots Leather Boots Observation: Auto Arrange not observed. Order changed after reshuffling.
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3. Temporary Interpretation (Observation-Based)

Current observations suggest:

- Auto Arrange may remain as an interference candidate only when existing arrangement candidates are insufficient.
- When enough arrangement candidates already exist, Auto Arrange appears less likely to remain.

This interpretation is observation-based and requires additional verification.

4. Summary Comparison Table

Condition A No existing arrangement Auto Arrange Leather Boots Formula A Auto Arrange observed	Condition B One existing arrangement Leather Boots Leather Boots Formula A Auto Arrange not observed	Condition C Two existing arrangements Formula A Leather Boots Leather Boots Auto Arrange not observed
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5. Observation Summary

- Auto Arrange no longer interference candidate when candidate sets are insufficient.
- Existing arrangement candidates may reduce Auto Arrange participation.
- Candidate count may influence Auto Arrange priority.
- Additional observations are required.

6. Important Notice

Observation-Based Research Model
This has extensive errors.
The actual current game algorithm has
not been implemented in the model under
actual circumstances.
Avoid interpreting this model as
an actual game rule.

 This figure is part of an observation-based research report, and
Results may vary depending on observation methods, equipment, and versions.

The source figure preserves the tested recipes, required-material differences, comparison lanes, and RF4SP/RF5 observation questions.

Interpret only within the displayed test conditions. Cross-title equivalence, trigger threshold, and internal selection remain Unknown.

SOURCE DETAIL + CURRENT BOUNDARY

06-09 Figure Restoration

Evidence boundary: the figure separates Observation, Model, and Unknown Transition as applicable.

Comparison point	RF4SP	RF5
Auto-Arrange effect	Partly unresolved	Candidate entry observed when slots are open
Visibility of boundary	Less distinct	The 3-to-4 boundary is easier to observe

6-5. Reselection When the Candidate Group Exceeds Three

One-line summary

In RF5, candidate competition may become visible when Candidate Count reaches four or more.

RF5 results that were relatively stable at three or fewer candidates became less stable after the group reached four or more through performance-source contamination, internal candidate generation, Auto-Arrange, or internal arrangements from the source equipment.

Observed changes include:

- different selected results;
- changes in displayed order;
- apparent displacement of a candidate; and
- failure of an Auto-Arrange candidate to appear.

For convenience, this document calls the observation **reselection after exceeding three candidates**. This label does not determine whether the internal process is a new random draw, candidate reduction, or displacement produced by priority handling.

The RF4SP 50-trial result in Section 6-2 shows that candidate-subset selection may be strongly uneven under at least one tested condition. It must not be generalized to all candidate-overflow conditions.

Comparison boundary: Messhilite 400-trial test

The RF4SP Messhilite test of candidate selection under overflow is a separate observation package reserved for Chapter 8. Across 400 trials, no comparable selection bias has been confirmed at present.

Therefore:

- the 300-trial Object X test concerns **final position**;
- the 50-trial turnip-material test concerns **which candidate subset remains**; and
- the 400-trial Messhilite test also concerns **candidate selection under overflow**, but did not show the same bias in the current analysis.

The difference is evidence against treating “RF4SP selection is weighted” as a general rule.

6-6. Boundary-Value Tests

One-line summary

Selection behavior and apparent candidate priority may change near the three-to-four-candidate boundary.

Boundary condition	Observed behavior	Provisional interpretation	Limitation
Candidate Count 1-3	Relatively fixed in RF5	Small groups may produce stable results	Most visible in RF5
Candidate Count 4+	More result and order variation	Candidate competition becomes visible	Internal process unresolved
Open arrangement slot	Auto-Arrange candidate observed	It may fill an otherwise open slot	Condition A-type observation
Existing arrangement slots	Auto-Arrange candidate not observed	Existing entries may take priority	Condition B-type observation
Existing entries produce four candidates	Order changes; Auto-Arrange not observed	Candidate competition may occur with lower Auto-Arrange	Condition C-type observation

Figure 06-P14-img-008

Master: PSC-CROSS-GAME-REV | Source anchor: Japanese Canonical p.14



Placement caption

RF4SP / RF5 disorder and display-order comparison
Evidence boundary: the figure separates Observation, Model, and Unknown Transition as applicable.

Boundary condition	Observed behavior	Provisional interpretation	Limitation
		priority	
RF4SP	Order variation from an earlier stage	Not the same clear 3-to-4 boundary as RF5	More trials needed

Candidate Count alone may not determine the outcome. Existing arrangements, Auto-Arrange, candidate type, and an unresolved priority or generation-order effect may also matter. Candidate maximum, fixed priority, and the exact threshold remain unresolved.

The additional RF4SP observations strengthen this boundary statement: a change in outcome can reflect either **subset selection** or **final ordering**, and the two must be tested separately.

6-7. The Order-Determination Problem

One-line summary

The final displayed order of arrangement entries may not be fixed by insertion order alone. Candidate-order changes and displayed-order changes occur even under conditions believed to contain the same candidate group. The final displayed order may therefore involve post-selection shuffling, priority handling, or displayed-order redetermination.

The **order in which candidates are selected** and the **order in which the selected candidates are displayed** may themselves be separate processes. The Object X 300-trial result provides direct post-Canonical evidence that final positions are not equally distributed under one RF4SP condition. It does not reveal which stage produced the bias.

Unresolved questions

1. Is the Object X position bias specific to Object X, to duplicated materials, or to candidate-generation order?
2. Does the 50-trial subset bias reproduce with different sets of four distinct candidates?
3. Does registration order or a material-specific property affect candidate selection?
4. Are subset selection and final displayed order separate processes?
5. Why does the 400-trial Messhilite overflow test not show the bias observed in the turnip-material test?
6. Do RF4SP and RF5 apply different post-generation processing?

The current evidence establishes differences in observed distributions. It does not identify the internal algorithm that creates them.

Source and update boundary

Sections 6-1 through 6-7 translate and reorganize the Japanese Canonical document 06_ .pdf. The Object X 300-trial test, the four-candidate 50-trial test, and the Messhilite comparison boundary are user-authorized post-Canonical additions dated 2026-08-08. They are based on gameplay observations and linked research sheets. They do not retroactively alter the historical Japanese Canonical source and do not claim to prove the game's internal implementation.

Figure 06-P17-img-010

Master: PSC-CROSS-GAME-REV | Source anchor: Japanese Canonical p.17

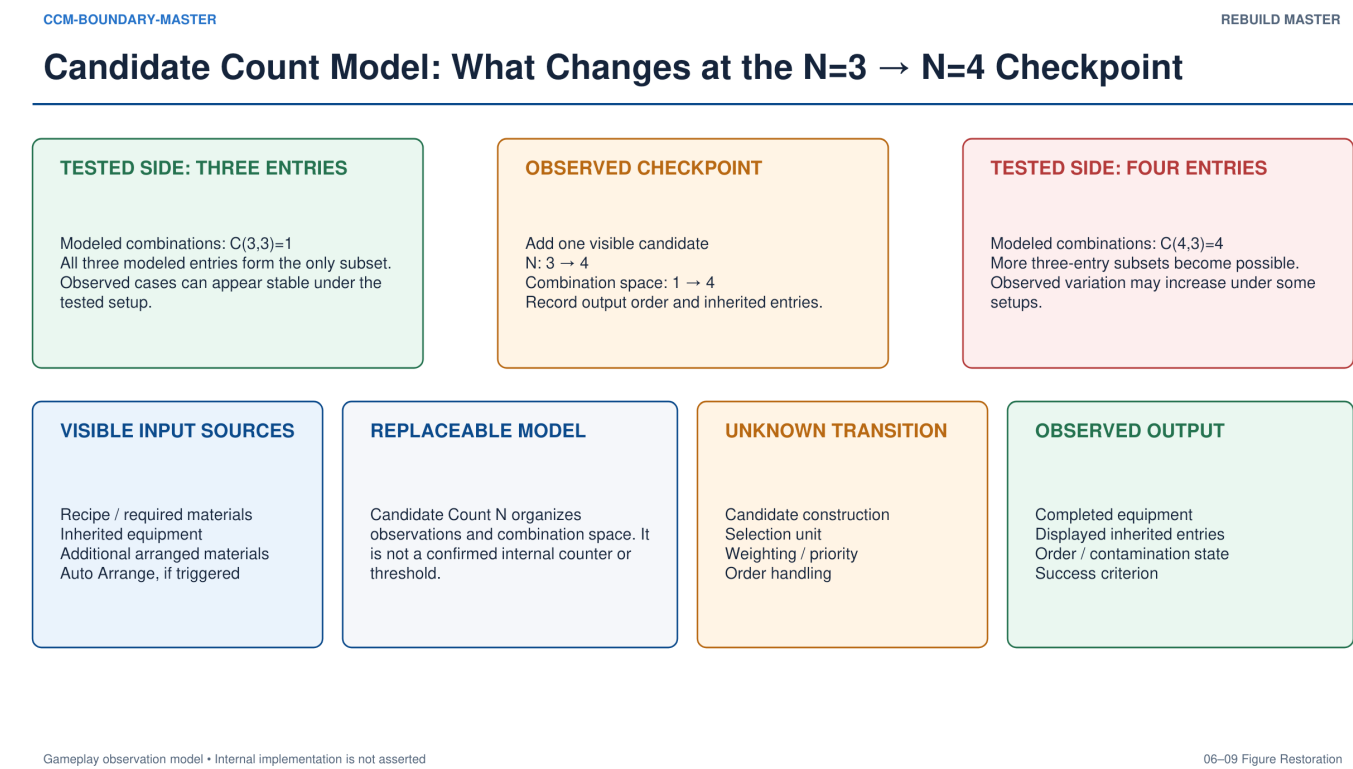


Placement caption

Comparison conditions for auto-arrange behavior
Evidence boundary: the figure separates Observation, Model, and Unknown Transition as applicable.

Figure 06-P18-img-011

Master: CCM-BOUNDARY-MASTER | Source anchor: Japanese Canonical p.18

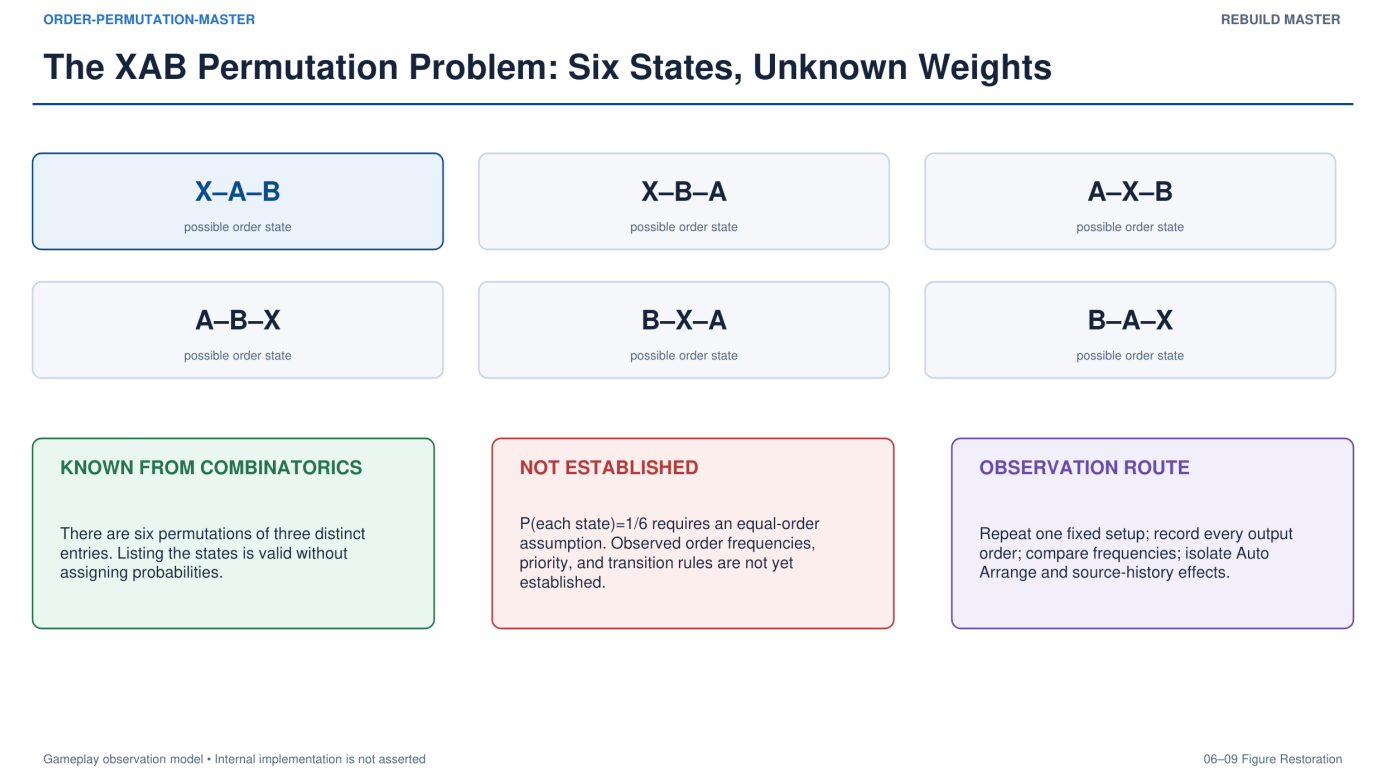


Placement caption

N=3 to N=4 boundary marker
Evidence boundary: the figure separates Observation, Model, and Unknown Transition as applicable.

Figure 06-P21-img-013

Master: ORDER-PERMUTATION-MASTER | Source anchor: Japanese Canonical p.21



Placement caption

RF4SP order-permutation possibility diagram
Evidence boundary: the figure separates Observation, Model, and Unknown Transition as applicable.